General Theory Of Relativity and Cosmology

Lecture Notes

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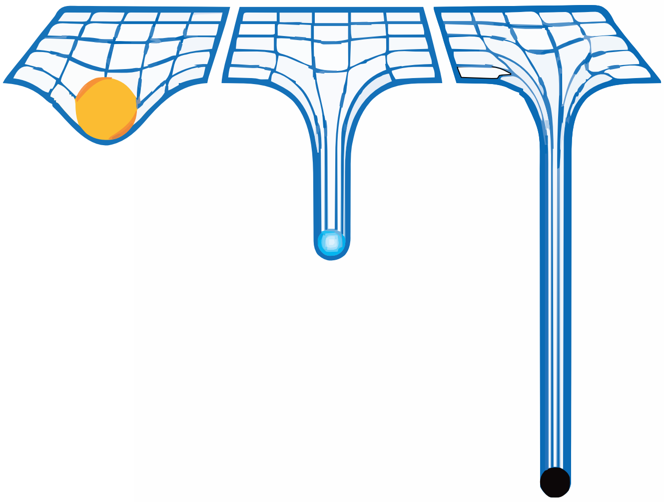

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Lecture 01: Introduction

In the physical sense, the word 'relativity' can be used to relate the measurement of one observer with that of another, for the same object/quantity. So using relativity, we can characterise the difference in the observations of two observers in two different frame of reference.

Note that, physical laws should not matter based on who is observing (that is, physical laws are *universal*), hence relativity becomes very important when dealing with the resolution of conflicts.

Let us consider two reference frames, and let v be the relative velocity between the frames.

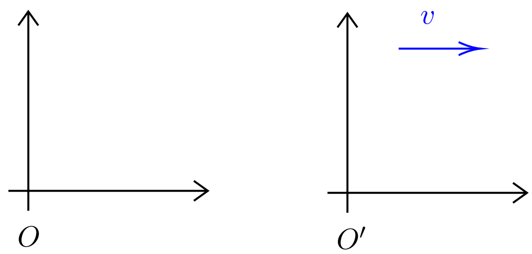
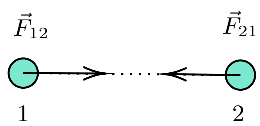


Figure 1: Two frames of reference with a relative velocity v

Fig. 1 shows two reference frames. The moving frame is denoted by O' (henceforth, primes will generally denote moving frames). If the relative velocity v between the two frames remain constant, then it falls under the domain of *Special Relativity* and if unfortunately (or fortunately) it doesn't, then it comes under *General Relativity*.

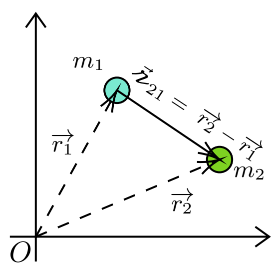
The theory of relativity dates back to the inception of Newtonian mechanics and hence, let us review the laws in brief (without being much technical 😅):

- **The First Law** :≈ velocity of a body remains constant unless acted by an external, unbalanced force.
- **The Second Law** :≈ $\vec{F} = m\vec{a}$ where $\vec{F}$ is the external force and $\vec{a}$ is the acceleration.
- **The Third Law** :≈ Every action has an equal and opposite reaction. Note that for this, two bodies are needed.



From the figure above, we will have $\vec{F}_{21} = -\vec{F}_{12}$

- **The Gravitational Law** :≈



In the above situation, if we consider for mass m_1, we have:

$$\vec{F}_{12} = G\frac{m_1 m_2}{\mathscr{r}_{21}^2}\hat{\mathscr{r}}_{21}$$

An important observation: in the second law, if we take the force to be zero, then apparently $\vec{a} = \frac{\mathrm{d}v}{\mathrm{d}t} = 0 \implies \vec{v} = \mathrm{const.}$ which is the statement of the first law. However, the second law does not imply the first law, since the first law defines what an *inertial frame* is and the statement of the second law applies only in case of inertial frames. So, a better formulation of the second and third law can be like, "In a frame where the first law is valid, blah blah blah..."

Lecture 02: Galilean Stuffs

In the previous lecture, we saw how Newton's laws define an inertial frame and how we can rephrase the second and third law in terms of intertial frames, to avoid confusion. It happens that, there are reference frames where the first law doesn't hold true.

Imagine a person inside a darkened car, isolated from the outside world, with a ball hanging from the roof of the car. The car suddenly starts accelerating and (as intuition speaks) the ball starts moving. The person claims "The ball has moved!" 💀.

Note that from the perspective of the person, no force has been applied to the ball, yet it moved, which is a direct violation of the second law. Thus, we can call this frame *non-inertial*.

Now, consider a person inside an ill-fated lift, tragically falling 🥲. Everything in the lift frame is accelerating together under gravity and hence 'weightlessness' (free-fall) occurs. If somehow a coin is also there in the lift, it simply floats. And if the person pushes the coin, it moves with an almost constant velocity. Thus, it is a very good approximation to an *intertial frame* (although, extremely tragic for the person!).

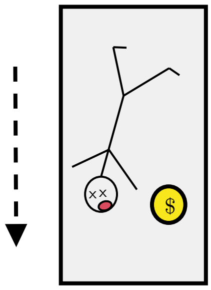


Figure 2: A sad guy floating with a coin in a falling lift

We had always wanted the physical laws to stay the same. This is actually the statement of Galilean relativity:

"Laws of physics (nature) must be the same in all intertial frames of reference"

By physical laws, we imply Newton's laws of mechanics and the law of gravitation. To have a more mathematical (and less philosophical) aspect to the above statement, we define the *Galilean transformation*.

2.1 Galilean Transformation

Let an onject be at point P and consider two reference frames O and O', which coincided at $t = 0$ but have moved apart since, along the common x-axis. (Henceforth, reference frames will simply be termed *observers*).

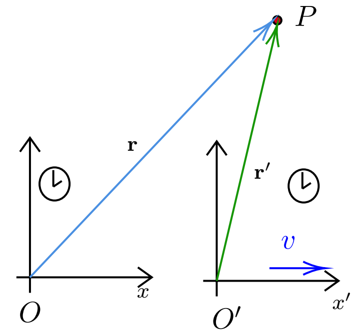


Figure 3: Observing same point from two different frames

Now, the distance between the origins of two frames, at time t, will be $R = vt$. Then accordingly, we can write the relation between the coordinates of the two observers as follows:

$$\begin{aligned} x' &= x - vt \\ y' &= y \\ z' &= z \\ t' &= t \end{aligned} \qquad \begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ -v & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix}$$

An important assumption: The Galilean transformation assumes a *universal time*, that is, time elapsed on clocks in both frames are the same [1].

Now, it is not necessary for the observers to move along the x axis only. Hence, generalising the transformation to arbitrary direction, we obtain:

$$t' = t \qquad \mathbf{r}' = \mathbf{r} - \mathbf{v}t$$

Now, let us look how the physical laws react under the Galilean transformation.

- **Law of Gravitation:**

  Consider the situation in the diagram below:

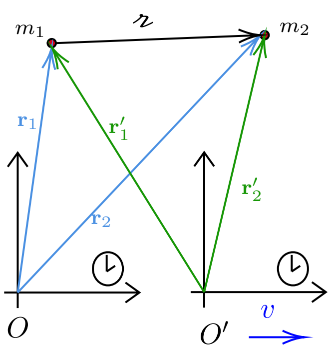


Figure 4: Gravitational law from two different frames

According to Galilean transformation, we have:

$$\begin{aligned} \boldsymbol{\mathscr{r}} &= \mathbf{r}_2 - \mathbf{r}_1 \\ &= (\mathbf{r}'_2 + \mathbf{v}t) - (\mathbf{r}'_1 + \mathbf{v}t) \\ &= \mathbf{r}'_2 - \mathbf{r}'_1 \\ &= \boldsymbol{\mathscr{r}}' \end{aligned}$$

[1] This was perhaps a fair assumption since time, atleast in the philosophical aspect, has always seemed to be *superior*.

Thus, we see that the vector $\mathscr{r}$ is unchanged and since this is the only vector appearing in Newton's law of gravitation, we have:

$$\mathbf{F}_{12} = G\frac{m_1 m_2}{\mathscr{r}^2}\ \hat{\mathscr{r}} = G\frac{m_1 m_2}{\mathscr{r}'^2}\ \hat{\mathscr{r}}' = \mathbf{F}'_{12}$$

We see that the expression of the force does not change between frames, thus indicating a *universality*.

- **Second Law:**

  Using the second law, we can write

  $$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$$

  Now, using Galilean transformation, we obtain:

  $$\mathbf{r}(t) = \mathbf{r}'(t') + \mathbf{v}t \implies \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \frac{\mathrm{d}\mathbf{r}'}{\mathrm{d}t} \times \frac{\mathrm{d}t'}{\mathrm{d}t} + \mathbf{v} \implies m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = m\frac{\mathrm{d}^2\mathbf{r}'}{\mathrm{d}t^2}$$

  Thus, we see that in general, force will be the same for observers in different inertial frames. Note that we assumed the *time universality*, that is, $\frac{\mathrm{d}t'}{\mathrm{d}t} = 1$

- **Third Law:**

  Note that since the force expression for gravity is symmetric in m_1 and m_2, the law of gravitation automatically incorporates the third law. Newton knew only about gravity and perhaps, third law would not have been necessary if he had relied only on gravity, however, he postulated that for all other forces, third law holds.

2.2 A Big Blow to Galileo

Let us consider that in Fig. 3, point P is also moving with some velocity $\mathbf{u}'$ and $\mathbf{u}$ in primed and unprimed frame respectively. Then, we can write:

$$\frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\big(\mathbf{r}' + \mathbf{v}t\big) = \frac{\mathrm{d}\mathbf{r}'}{\mathrm{d}t} + \mathbf{v} = \mathbf{u}' + \mathbf{v} \implies \mathbf{u} = \mathbf{u}' + \mathbf{v} \qquad : \text{velocity addition formula}$$

However, observations, specially the Michelson-Morley experiment, were made in different frames but found the speed of light (henceforth denoted as c) to be a constant, which contradicted the velocity addition formula when applied to speed of light.

Since the velocity addition formula depended entirely on the Galilean transformation, to resolve this conflict, we have to change the transformation rule in its entirety.

Lecture 04: Vectors and Stuffs

Let us consider rotation about the z-axis by an angle θ, in which case, the x and y components transform as:

$$x' = x\cos\theta + y\sin\theta$$
$$y' = -x\sin\theta + y\cos\theta$$

Then, we can write the infinitesimal change in the components in the following form:

$$\begin{pmatrix} dx' \\ dy' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}$$

Note that, this can also be written as:

$$\begin{pmatrix} dx' \\ dy' \end{pmatrix} = \begin{pmatrix} \frac{\partial x'}{\partial x} & \frac{\partial x'}{\partial y} \\ \frac{\partial y'}{\partial x} & \frac{\partial y'}{\partial y} \end{pmatrix} \begin{pmatrix} dx \\ dy \end{pmatrix}$$

The matrix in this case is the *Jacobian matrix* and this is a more general structure of coordinate transformations and hence any transformation can be written like this, even the Galilean or the Lorentz transformations.

This shows that transformations are just multiplication of a matrix with a vector. To this extent, we define a *vector* [1] $\bar{x}$ with the following components:

$$\bar{\mathbf{x}} = (x^0, x^1, x^2, x^3)^T$$

Here $x^0 = ct$ and $x^1 = x, x^2 = y, x^3 = z$. Note that since t and x are not dimensionally same, we took $x^0 = ct$ to make it in terms of ditance. We multiplied by c (and not any other velocity) because c is a universal constant.

We can express any infinitesimal transformation as:

$$\begin{pmatrix} dx'^0 \\ dx'^1 \\ dx'^2 \\ dx'^3 \end{pmatrix} = \begin{pmatrix} \frac{\partial x'^0}{\partial x^0} & \cdots & \frac{\partial x'^0}{\partial x^3} \\ \vdots & \ddots & \vdots \\ \frac{\partial x'^3}{\partial x^0} & \cdots & \frac{\partial x'^3}{\partial x^3} \end{pmatrix} \begin{pmatrix} dx^0 \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix}$$

In a more compact way, we can write in the vector-matrix form or in the component form using Einstein's summation convention:

$$\mathbf{d\bar{x}}' = \Lambda \mathbf{d\bar{x}} \iff dx'^\mu = \Lambda^\mu{}_\nu dx^\nu$$

where $\Lambda^\mu{}_\nu = \frac{\partial x'^\mu}{\partial x^\nu}$ is an element of the *matrix* Λ.

Summation convention: We drop the summation sign if twice repeated indices are there on the same side of the equation.

Now, let us define what a vector is:

Definition 1 (Vector):
If a mathematical object transforms under a coordinate transformation like a differential $\mathbf{d\bar{x}}$, that is, $\mathbf{d\bar{x}} \to \mathbf{d\bar{x}}' = \Lambda\, \mathbf{d\bar{x}}$, then it is called a *contravariant vector* or a *contravector* or a *tensor of rank* $(1, 0)$

Let us show that under the Galilean transformation, velocity, as defined by $\mathbf{u} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$ is a vector. We will define everything in contravariant and four-vector notation. For this, we consider the vector defined as before:

$$\mathbf{d\bar{x}} = (ct, x, y, z)^T \equiv (x^0, x^1, x^2, x^3)^T$$

The Galilean transformation is defined by:

$$\lambda \equiv \begin{pmatrix} 1 & 0 & 0 & 0 \\ -^{v^1}/_c & 1 & 0 & 0 \\ -^{v^2}/_c & 0 & 1 & 0 \\ -^{v^3}/_c & 0 & 0 & 1 \end{pmatrix} \iff \begin{aligned} x'^0 &= x^0 \\ x'^1 &= x^1 - (^{v^1}/_c)x^0 \\ x'^2 &= x^2 - (^{v^2}/_c)x^0 \\ x'^3 &= x^3 - (^{v^3}/_c)x^0 \end{aligned}$$

[1] This is often called a *four* vector

We can then define the four velocity as[1]:

$$u^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}t} = (c, u^1, u^2, u^3)^T$$

Now, we have:

$$\begin{aligned} dx'^\mu &= \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \implies \quad u'^\mu \cancel{dt'} &= \frac{\partial x'^\mu}{\partial x^\nu} u^i \cancel{dt} \qquad \text{(as } dt = dt'\text{)} \\ \implies \quad u'^\mu &= \frac{\partial x'^\mu}{\partial x^\nu} u^\nu \end{aligned}$$

Thus, we see that the components of the velocity vector transform similar to the differential displacement and hence, velocity is indeed a vector under Galilean transformation. In a similar way, we can show that acceleration $\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}$ is also a vector under Galilean transformation.

Now, let us turn to Lorentz transformation. Things become bad here! For example, consider the same definition of velocity $\mathbf{u} = \frac{\mathrm{d}\bar{\mathbf{x}}}{\mathrm{d}t}$. However, here t does not remain invariant under coordinate transformation. Then we will have:

$$u'^\mu = \frac{\mathrm{d}x'^\mu}{\mathrm{d}t'} = \frac{\Lambda^\mu{}_\nu x^\nu}{\Lambda^0{}_\alpha dx^\alpha} \neq \frac{\mathrm{d}x^\mu}{\mathrm{d}t}$$

Thus, this definition fails to transform similar to a differential displacement and hence, the way it is defined presently, velocity is not a vector 😭.

Lecture 05: Some more Vectors and Stuffs

Previously, we saw how a vector was defined in terms of the transformation of the differential displacements. In here, we will consider another situation.

First, note that, using a given transformation,

$$dx'^\mu = \Lambda^\mu{}_\nu dx^\nu$$

we can define the inverse transformation,

$$dx^\nu = (\Lambda^{-1})^\nu{}_\mu dx'^\mu$$

The above statement is valid only if the transformation matrix is invertible (in most case, transformations like translations, rotations, etc. are invertible).
Let us now see what the inverse transformation matrix looks like. For that, consider:

$$\begin{aligned} dx'^\mu &= \frac{\partial x'^\mu}{\partial x^\nu} dx^\nu \\ \implies \frac{\partial x^\beta}{\partial x'^\mu} dx'^\mu &= \underbrace{\frac{\partial x^\beta}{\partial x'^\mu}\frac{\partial x'^\mu}{\partial x^\nu}}_{\delta^\beta{}_\nu} dx^\nu \\ \implies \qquad dx^\beta &= \frac{\partial x^\beta}{\partial x'^\mu} dx'^\mu \end{aligned}$$

Thus we see that:

$$(\Lambda^{-1})^\nu{}_\mu \equiv \frac{\partial x^\nu}{\partial x'^\mu}$$

[1]Note that four velocity is defined as derivative with respect to something called as *proper time*, however, for Galilean transformations, proper time coincides with the coordinate time and hence this definition is okay.

5.1 Gradient Vector

In the Cartesian coordinates, we all know:

$$\boldsymbol{\nabla} = \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}$$

In line with this, we will define another quantity (often called *four gradient*) such that:

$$\nabla_\mu := \frac{\partial}{\partial x^\mu}$$

A few points on this:

1. Note that we wrote the index in the down position for ∇_μ. This is done since we will see that the four gradient does not follow the transformation rule of a vector, hence it is a different entity. Since in vectors we put the indices upwards, to differentiate it from a vector, we put it down.

2. In spherical or cylindrical or any other coordinates, the *ordinary* gradient is not just the partial derivatives of the coordinates. There are some obnoxious prefactors involved also. Irrespective of that, we define the four gradient as just the derivative of the coordinates ('coz later we will see that those prefactors are just some jugglery with the *metric*).

The four gradient is better denoted by ∂_μ and thus, we have:

$$\bar{\boldsymbol{\partial}} \equiv \begin{pmatrix}\partial_0 & \partial_1 & \partial_2 & \partial_3\end{pmatrix} = \begin{pmatrix}\frac{1}{c}\frac{\partial}{\partial t} & \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\end{pmatrix}$$

Under a coordinate transformation $x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu$, we have using chain rule:

$$\partial'_\mu = \frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial}{\partial x^\nu} \implies \partial'_\mu = \left(\Lambda^{-1}\right)^\nu{}_\mu \frac{\partial}{\partial x^\nu}$$

The four gradient apparently transform accordingly as $\bar{\boldsymbol{\partial}}' = \Lambda^{-1}\boldsymbol{\partial}$ which is contrary to that of the vector.

> **Definition 2:**
> Covector If a mathematical object transforms under a coordinate transformation like the derivative $\bar{\boldsymbol{\partial}}$, then we call it a *covariant vector* or *covector* or a *tensor of rank (0,1)*

Actually vectors and covectors are defined based how the basis vectors of the space transform. Vectors transform in a way *contrary* to the basis vectors and hence are called *contravariant* while covectors transform in a way similar to how the basis transforms and hence called *covariant*.[1]

5.2 Tensors

Consider the product,

$$\boldsymbol{\omega} = \bar{\mathbf{x}}\bar{\mathbf{x}}^T \equiv \begin{pmatrix} dx^0 \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix} \begin{pmatrix} dx^0 & dx^1 & dx^2 & dx^3 \end{pmatrix} = \begin{pmatrix} dx^0 dx^0 & \dots & dx^0 dx^3 \\ \vdots & \ddots & \vdots \\ dx^3 dx^0 & \dots & dx^3 dx^3 \end{pmatrix}$$

Note that, defined like this, $\boldsymbol{\omega}$ represents a square matrix and thus, has two indices. This operation is often called *outer product* and is denoted as

$$\boldsymbol{\omega} = \bar{\mathbf{x}} \otimes \bar{\mathbf{x}} \qquad \omega^{\mu\nu} = dx^i dx^j$$

[1] In the elegant language of differential geometry, vectors are actually the *tangent vectors* belonging to the *tangent space* of the space (*manifold*) while covectors belong to the *dual space* of the *tangent space*.

Now, let us see how it transforms:

$$\begin{aligned}
\omega^{\mu\nu} &= dx'^{\mu}dx'^{\nu} \\
&= \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}}dx^{\alpha}\right)\left(\frac{\partial x'^{\nu}}{\partial x^{\beta}}dx^{\beta}\right) \\
&= \left(\frac{\partial x'^{\mu}}{\partial x^{\alpha}}\right)\left(\frac{\partial x'^{\nu}}{\partial x^{\beta}}\right)dx^{\alpha}dx^{\beta}
\end{aligned}$$

This is yet another kind of transformation, different from either vector and covector.

Definition 3 (Tensor):
If a mathematical object transforms, under a coordinate transformation, like $\underbrace{\mathbf{d\bar{x}}\otimes\ldots\otimes\mathbf{d\bar{x}}}_{m\text{ times}}$ and like $\underbrace{\bar{\boldsymbol{\partial}}\otimes\ldots\otimes\bar{\boldsymbol{\partial}}}_{n\text{ times}}$, then it is called a *tensor of rank (m,n)*

A prime example would be the *moment of inertia tensor* I^{ij} or the *electromagnetic tensor* $F_{\mu\nu}$. We note that tensors have upstairs indices for each contravariant components and downstairs indices for each covariant component. [1].

Definition 4 (Scalar):
If a mathematical object does not transform under a coordinate transformation, then it is called a *scalar* or a *tensor of rank (0,0)*.

Few example of scalars would be the invariant displacement $dr^2 = dx^2 + dy^2 + dz^2$ under Galilean transformation and the invariant spacetime interval $ds^2 = -c^2dt^2 + dr^2$ under Lorentz transformation.

5.3 Inner Product

Previously we saw the *outer product* and we expect an *inner product* too. Multiplying a vector with the transpose changed it to a square matrix. Here, we do the opposite. Consider the scalar dr^2 under Galilean transformation, which is written as:

$$dr^2 = \mathbf{dr}\cdot\mathbf{dr} = \mathbf{dr}^{\mathrm{T}}\mathbf{dr} = \begin{pmatrix} dx^1 & dx^2 & dx^3 \end{pmatrix}\begin{pmatrix} dx^1 \\ dx^2 \\ dx^3 \end{pmatrix}$$

However, notice the apparent problem if we generalise this to Lorentz transformation too. We have:

$$ds^2 = -c^2dt^2 + dr^2$$

This cannot be simply written as $\mathbf{d\bar{x}}^{\mathrm{T}}\mathbf{d\bar{x}}$, all due to that *innocent* minus sign in front of $c^2dt^2 \equiv (dx^0)^2$.

How do we generalise this notion of scalar as some matric product?

For that, let us define another matrix which will take care of the minus sign:

$$[g] \equiv \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$$

[1]We can easily change upstairs to downstairs indices by the *metric* and hence it is better to write tensors as ${T^{\mu}}_{\nu}$ and not T^{μ}_{ν} so as to keep a vacant place for the indices to move up/down.

Using this, we can check that:

$$ds^2 = \begin{pmatrix} dx^0 & dx^1 & dx^2 & dx^3 \end{pmatrix} \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} \begin{pmatrix} dx^0 \\ dx^1 \\ dx^2 \\ dx^3 \end{pmatrix} = \mathbf{d\bar{x}}^{\mathrm{T}} \, g \, \mathbf{d\bar{x}}$$

This *g* is called the *metric tensor* and in component form, the above can be written as:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

Note that, for Galilean transformation, the metric tensor is just the *identity* matrix,

$$[g]_{\mathrm{GT}} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix}$$

Lecture 06: Metric is also a Matrix

Previously, we saw that to get a scalar from a contra-vector, we had to use an additional matrix which led to the notion of the *metric tensor*. Let us analyse this object further...

6.1 Transformation of the metric

For this, we will consider the *sacred* rule that space-time interval should remain invariant under coordinate change. Thus, we have:

$$\begin{aligned} ds^2 &= ds'^2 \\ \implies \quad g_{\mu\nu} dx^\mu dx^\nu &= g'_{\alpha\beta} dx'^\alpha dx'^\beta \\ \implies g_{\mu\nu} \left(\frac{\partial x^\mu}{\partial x'^\alpha} dx'^\alpha \right) \left(\frac{\partial x^\nu}{\partial x'^\beta} dx'^\beta \right) &= g'_{\alpha\beta} dx'^\alpha dx'^\beta \\ \implies g_{\mu\nu} \left(\frac{\partial x^\mu}{\partial x'^\alpha} \right) \left(\frac{\partial x^\nu}{\partial x'^\beta} \right) dx'^\alpha dx'^\beta &= g'_{\alpha\beta} dx'^\alpha dx'^\beta \end{aligned}$$

And now by the eternal rule for arbitrariness (of the different displacements), we conclude that:

$$\boxed{g'_{\alpha\beta} = g_{\mu\nu} \left(\frac{\partial x^\mu}{\partial x'^\alpha} \right) \left(\frac{\partial x^\nu}{\partial x'^\beta} \right)}$$

Thus, we see that the metric tensor transforms as a rank (0,2) tensor and hence, we were not wrong about calling this a *tensor*!

Few Properties of the metric:

- Note the following :

  $$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = g_{\nu\mu} dx^\nu dx^\mu = g_{\nu\mu} dx^\mu dx^\nu$$

  In the first step, we interchanged $\mu \leftrightarrow \nu$ since these are dummy indices and then we interchanged the placement of $dx^\mu \leftrightarrow dx^\nu$ since these are just components and hence commutes. Then, from this, we have $g_{\mu\nu} = g_{\nu\mu}$ which leads us to the fact that:

Metric tensor is symmetric

- If we have g to be non-singular, that is, $\det(g) \neq 0$, then we can define an inverse metric such that:

  $$gg^{-1} = g^{-1}g = \mathbb{1}$$

  In the component form, we can write

  $$g_{\mu\nu}(g^{-1})^{\nu\alpha} = \delta^{\alpha}{}_{\mu}$$

  We will henceforth use $g^{\nu\alpha}$ instead of $(g^{-1})^{\nu\alpha}$ for the inverse metric, since this is more sleek visually and also, it does not confuse us.

Note that, by the same logic as we showed that metric is a tensor, we can also show that inverse metric is a rank (2,0) tensor.

6.2 Metric is a map!

The metric (and its inverse too) is a map which takes a vector (covector) and changes it to a covector (vector) according as:

$$\begin{aligned} A_\mu &\xmapsto{g} A^\nu & A_\mu &= g_{\mu\nu}A^\nu \\ A^\mu &\xmapsto{g^{-1}} A_\nu & A^\mu &= g^{\mu\nu}A_\nu \end{aligned}$$

Using this, the norm of a vector can be written as:

$$\left\|\bar{\mathbf{A}}\right\| = \bar{\mathbf{A}} \cdot \bar{\mathbf{A}} = A_\mu A^\mu = A^\mu A_\mu$$

Lecture 07: Metric and STR

Consider the following transformation:

$$(r,\theta) \mapsto (x,y) \qquad x = r\cos\theta \quad y = r\sin\theta$$

Let us calculate the transformation matrix for this transformation. Note that, here we have $\mathbf{x} \equiv (r,\theta)$ and $\mathbf{x}' \equiv (x,y)$. We calculate Λ from its definition as:

$$\begin{aligned} x &= r\cos\theta \\ y &= r\sin\theta \end{aligned} \qquad \Lambda \equiv \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

Once we have obtained the transformation matrix, we now move on to calculate the metrix tensor in polar coordinates. For that, note, for Cartesian coordinate,

$$ds^2 = dx^2 + dy^2 \implies g \equiv \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}$$

Then, according to this question, we already have the knowledge of $g'_{\mu\nu}$ and we have to transform it to $g\alpha\beta$ and we already know the transformation rule. Also, since g' is diagonal, we need to only consider the diagonal components. Thus, all of our work is already done!

$$g_{\alpha\beta} = \Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta}g'_{\mu\nu}$$

- $g_{rr} = \Lambda^{x}{}_{r}\Lambda^{x}{}_{r}g'_{xx} + \Lambda^{y}{}_{r}\Lambda^{y}{}_{r}g'_{yy} = \cos^2\theta + \sin^2\theta = 1$

- $g_{\theta\theta} = \Lambda^x{}_\theta \Lambda^x{}_\theta g'_{xx} + \Lambda^y{}_\theta \Lambda^y{}_\theta g'_{yy} = r^2(\sin^2\theta + \cos^2\theta) = r^2$
- $g_{r\theta} = \Lambda^x{}_r \Lambda^x{}_\theta g'_{xx} + \Lambda^y{}_r \Lambda^y{}_\theta g'_{yy} = -r\cos\theta\sin\theta + r\sin\theta\cos\theta = 0$ (hence this metric is diagonal too!)

Thus, from this, we obtain

$$g \equiv \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix}$$

From this, the invariant distance becomes our know result:

$$ds^2 = g_{ij}x^i x^j = dr^2 + r^2(d\theta)^2$$

This is a weirdly powerful technique to find the metric in any coordinate system. And once we have found the metric, then calculations of stuff like gradient and Laplacians become very easy.

7.1 Proper Time

Time no longer remains a scalar in special theory of relativity. However, we do need some notion of invariant time which can be useful to define some quantities. At present, we only know two scalars in STR, that is, c and ds^2. Note that, dimensionally, there is only one way to construct some 'time' using these two:

$$d\tau^2 := -\frac{ds^2}{c^2}$$

We dfine $d\tau$ to be the *proper time* which is a scalar, since it is constructed out of two scalars. The minus sign in the defnition results from the convention of the $(-,+,+,+)$ metric that we had chosen.

Consider two events $\epsilon_1 = (ct_1, x, y, z)$ and $\epsilon_2 = (ct_2, x, y, z)$, which are happening at the same spatial location. Then, since $dr^2 = 0$, for these two events,

$$ds^2 = -c^2 dt^2 \implies dt = d\tau$$

Hence the time interval between two events happening at the same spatial location is the proper time. Note that, using this notion of time (which is a scalar) we can define velocities and momentum.

7.2 Looking back at STR

Let us now define some quantities:

- **four-position:** denoting the position of an object in space-time

$$\bar{\mathbf{x}} = (x^0, x^1, x^2, x^3)^{\mathrm{T}}$$

- **four-velocity:** denoting a velocity of the object in space-time

$$\bar{\mathbf{u}} = \frac{\mathrm{d}\bar{x}}{\mathrm{d}\tau} = \begin{pmatrix} \frac{\mathrm{d}(ct)}{\mathrm{d}\tau} \\ \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}\tau} \end{pmatrix}$$

  Note that, under this defintion, $\bar{\mathbf{u}}$ becomes a tensor of rank $(1, 0)$.

  Now, consider two reference frames; the moving frame moves with a velocity $\mathbf{v}$ with respect to the lab frame. Consider an observer at rest in the moving frame. Then, in the observer's frame, since spatial coordinates remain same, we have $ds'^2 = -c^2 d\tau^2$ while for the lab frame, we have $ds^2 = -cdt^2 + dx^2 + dy^2 + dz^2 = -c^2dt^2 + dr^2$. Now, since the observer is at rest, they move with the same velocity $\mathbf{v}$ with respect to the lab frame and hence $\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{v}$. Hence, by the requirement that $ds'^2 = ds$ we have:

$$\begin{aligned} d\tau^2 &= -\frac{ds^2}{c^2} \\ &= -\frac{1}{c^2}\left(-c^2dt^2 + dr^2\right) \\ &= dt^2\left[1 - \frac{1}{c^2}\left(\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}\right)^2\right] \end{aligned}$$

From this we obtain that $d\tau = {}^{dt}/_{\gamma}$ where $\gamma := \frac{1}{\sqrt{1-(v^2/c^2)}}$. Substituting this in the above expression for four-velocity, we obtain:

$$u^\mu \equiv \gamma(c, {}^{d\mathbf{x}}/_{dt})^{\mathrm{T}} = \gamma(c, \mathbf{v})$$

- **four-momentum:** momentum of an object in space-time

  $$\bar{\mathbf{p}} = m_o\bar{\mathbf{u}} = m_o\gamma(c, \bar{\mathbf{v}}) = \gamma(m_oc, \mathbf{p})$$

  Note that m_o as here is simply the *mass*. The term rest-mass is often used, however, it often confuses us since this implies the existence of some *relativisitc mass*.

  Earlier, we used to define the relativistic mass as γm_o which kept the expression for spatial components of four-momentum and three momentum similar (that is, mass times velocity).

  However, in recent times, the concept of relativistic mass is being done away with and we instead now change the expression of momentum when considering STR.

7.2.1 Metric in STR

We have the invariant displacement as:

$$ds^2 = -c^2dt^2 + dr^2$$

From this, we infer that the metric $g = \mathrm{diag}(-1, 1, 1, 1)$. This metric is also termed as the *Minkowski metric*. Using this metric, we have:

$$\|\bar{\mathbf{u}}\|^2 = g_{\mu\nu}u^\mu u^\nu = -\gamma^2c^2 + \gamma^2\|\mathbf{v}\|^2 = \gamma^2c^2\left(\frac{\|v\|^2}{c} - 1\right) = \cancel{\gamma^2}c^2 \times \frac{-1}{\cancel{\gamma^2}} = -c^2$$

Thus, we obtain that the squared-norm of the four-velocity is always $-c^2$ 🤯, which is kinda cool since even without doing anything, we technically *move* at the speed of light!!

In a similar way, we obtain $\|\bar{\mathbf{p}}\|^2 = m_o^2\|\bar{u}\|^2 = -m_o^2c^2$